

Airline Passenger Satisfaction Prediction

Notebook link : <https://drive.google.com/file/d/1kgjmvxNOTWNMtL1Pfstd3FCg6d77bILG/view?usp=sharing>

Executive Summary

Objective

- Predict passenger satisfaction using demographic information, flight characteristics, service quality ratings, and delay-related features.

Dataset

- Airline passenger experience dataset.
- Mixed numerical and categorical variables describing customer profiles, travel conditions, and onboard services.
- Binary target variable: Satisfied / Dissatisfied.

Methodology

- Exploratory Data Analysis (EDA)
- Statistical Feature Selection
- Logistic Regression
- Random Forest
- AdaBoost
- Hyperparameter Tuning

Key Findings

- Service quality indicators and travel class are among the most influential predictors of satisfaction.
- Ensemble methods outperform the baseline and linear models.
- Random Forest achieved the best overall predictive performance.

Dataset Overview

Records

~130,000 passengers

Features

23 variables

Target

Satisfaction(Satisfied / Dissatisfied)

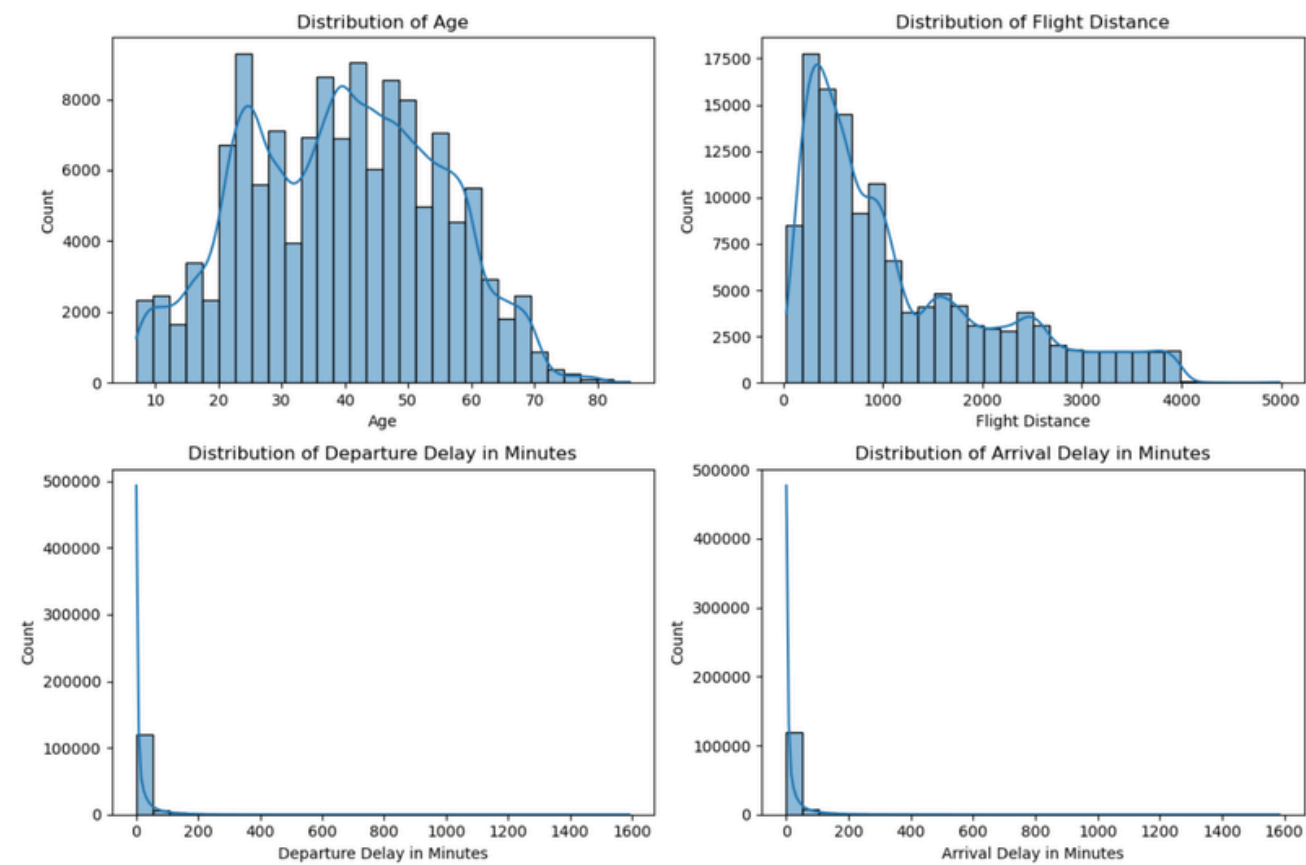
Feature Groups

- Demographics
- Travel Characteristics
- Service Ratings
- Delay Information

Numerical Feature Distributions

The numerical variables exhibit different distribution patterns.

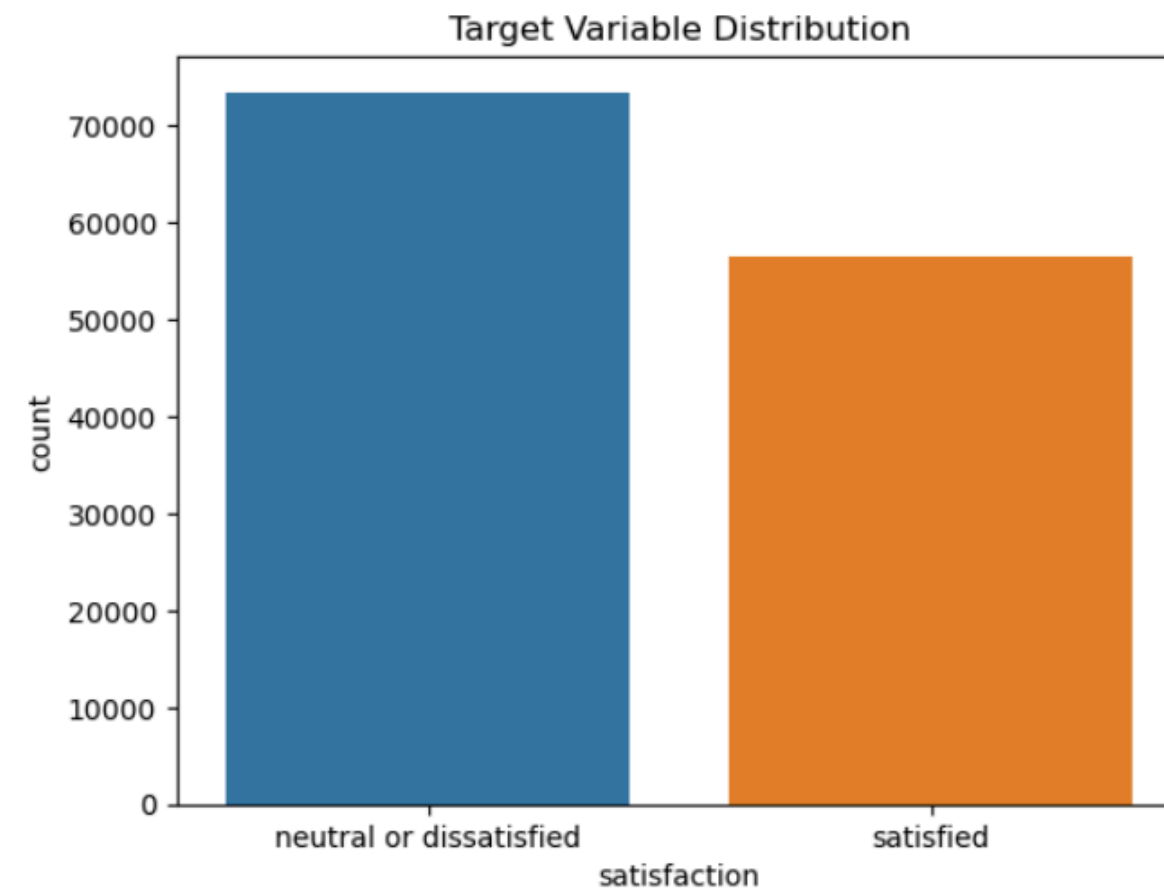
Age and Flight Distance show relatively broad distributions, while Departure Delay and Arrival Delay are strongly right-skewed. This indicates that most flights experience limited delays, whereas a smaller number of flights are affected by substantial disruptions.



Target Variable Distribution

The target variable is reasonably balanced, with approximately 57% of passengers classified as satisfied and 43% as dissatisfied.

This class distribution suggests that accuracy can be considered an appropriate evaluation metric, as severe class imbalance is not present.



Passenger Satisfaction by Customer Type

Customer type appears to have a strong relationship with satisfaction.

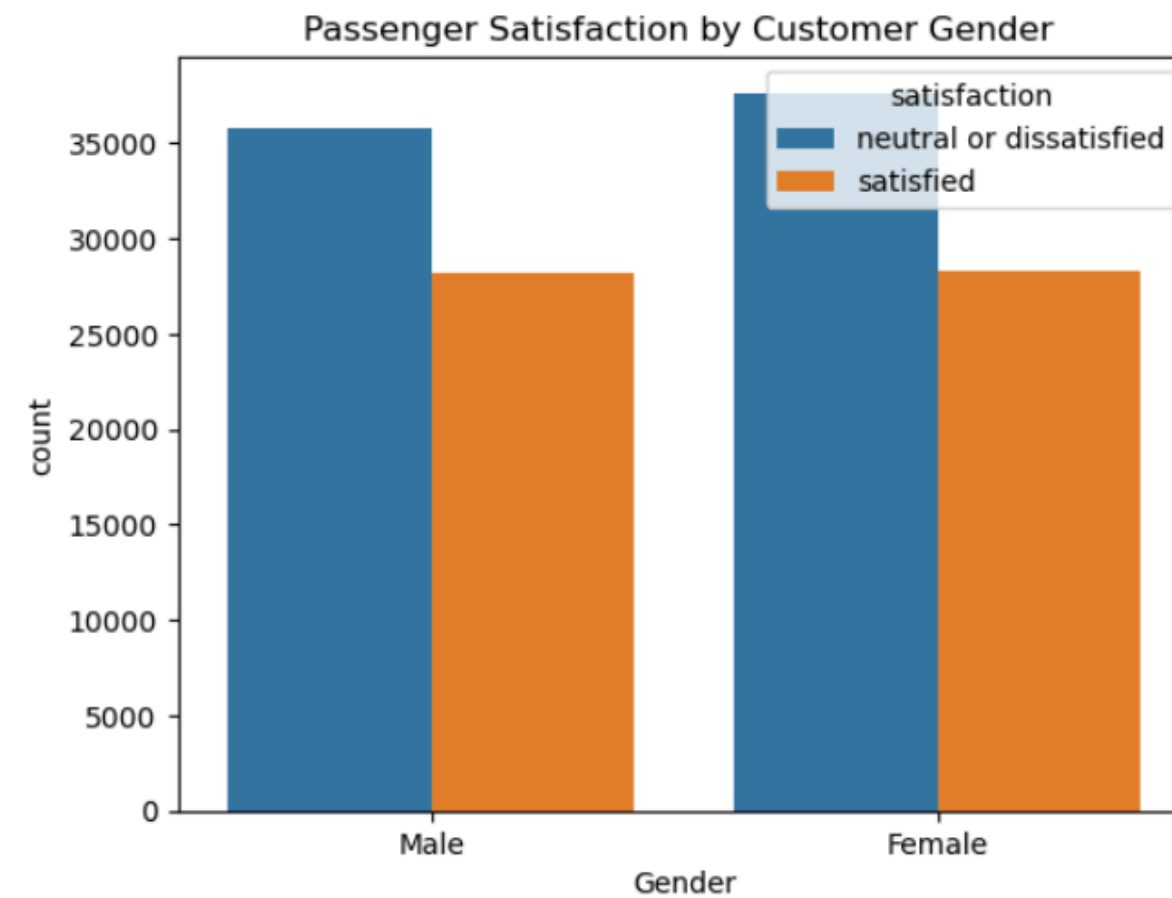
Loyal customers tend to report higher satisfaction levels, while disloyal customers are more frequently associated with dissatisfaction.



Passenger Satisfaction by Customer Gender

The distribution of satisfaction across genders is relatively similar.

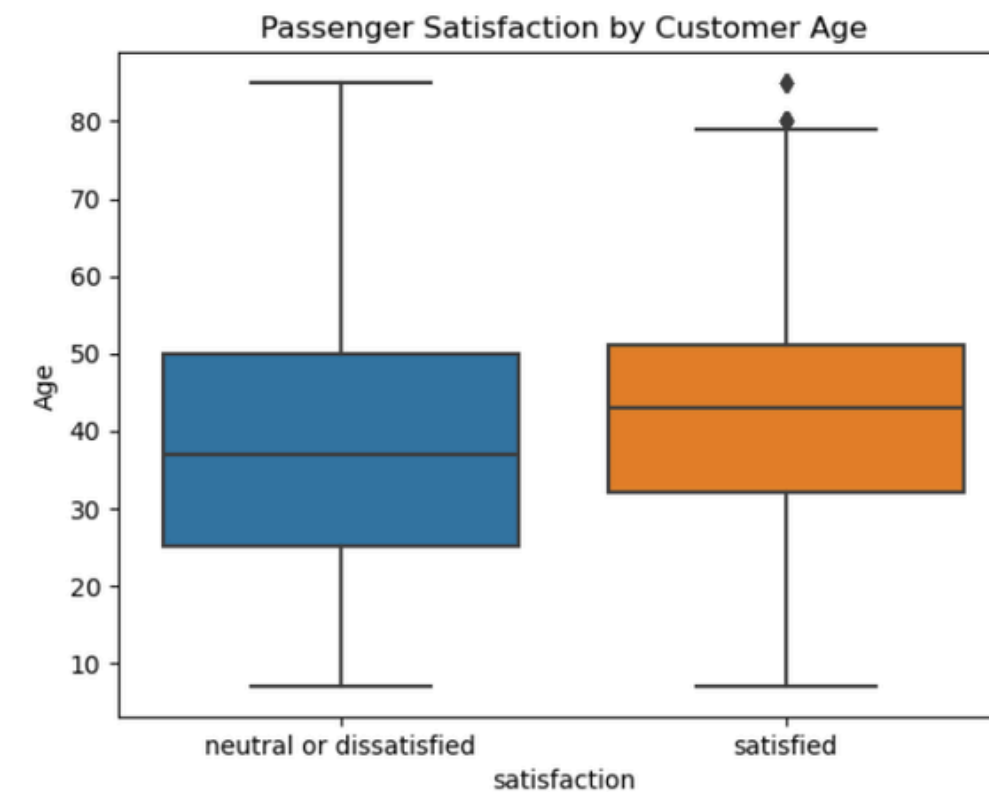
This suggests that gender alone is unlikely to be a strong predictor of passenger satisfaction.



Passenger Satisfaction by Customer Age

Passenger age shows variability across satisfaction groups.

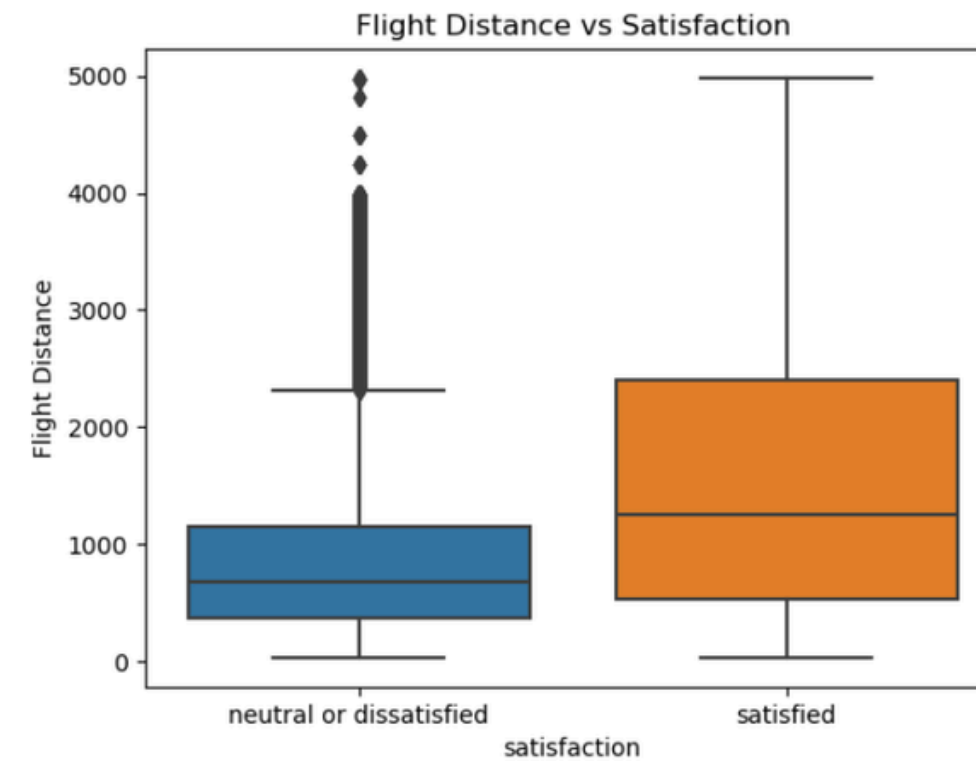
Although some differences are observable, age alone does not appear to provide a clear separation between satisfied and dissatisfied passengers.



Passenger Satisfaction by Flight Distance

Flight distance appears to be associated with passenger satisfaction.

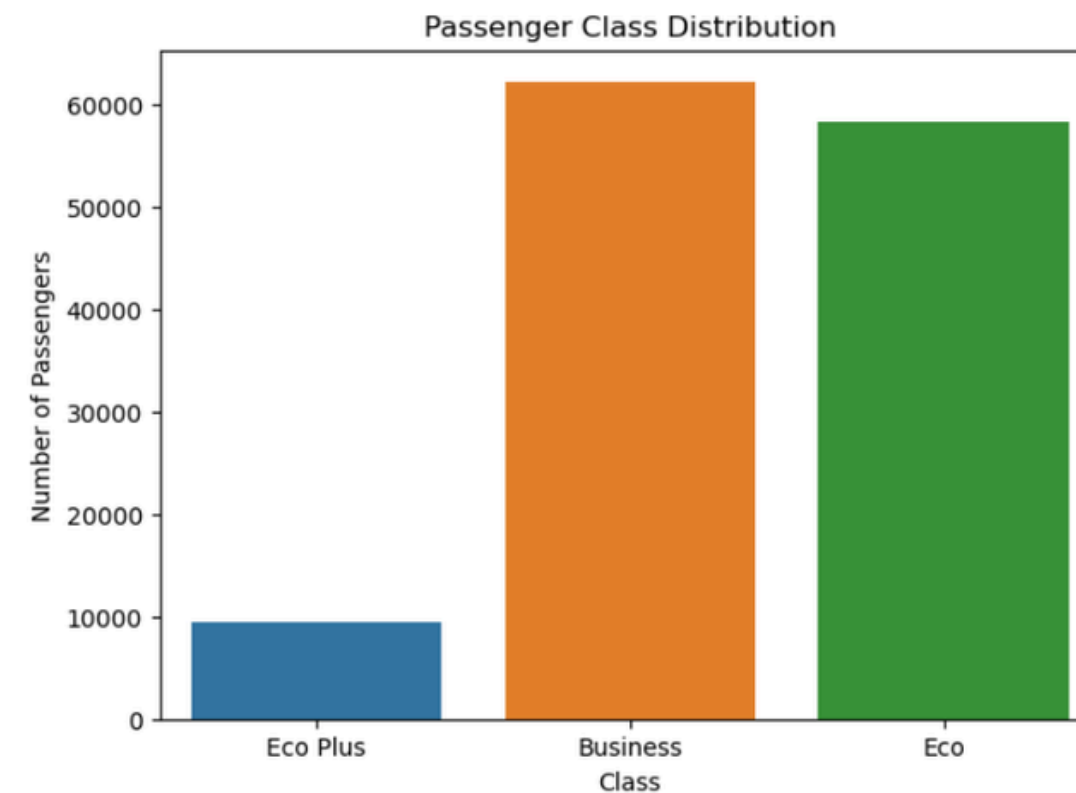
Satisfied passengers tend to travel longer distances on average, suggesting that flight length may influence the overall travel experience.



Passenger Class Distribution

Economy and Economy Plus are the most represented travel classes in the dataset, while Business Class accounts for a smaller proportion of passengers.

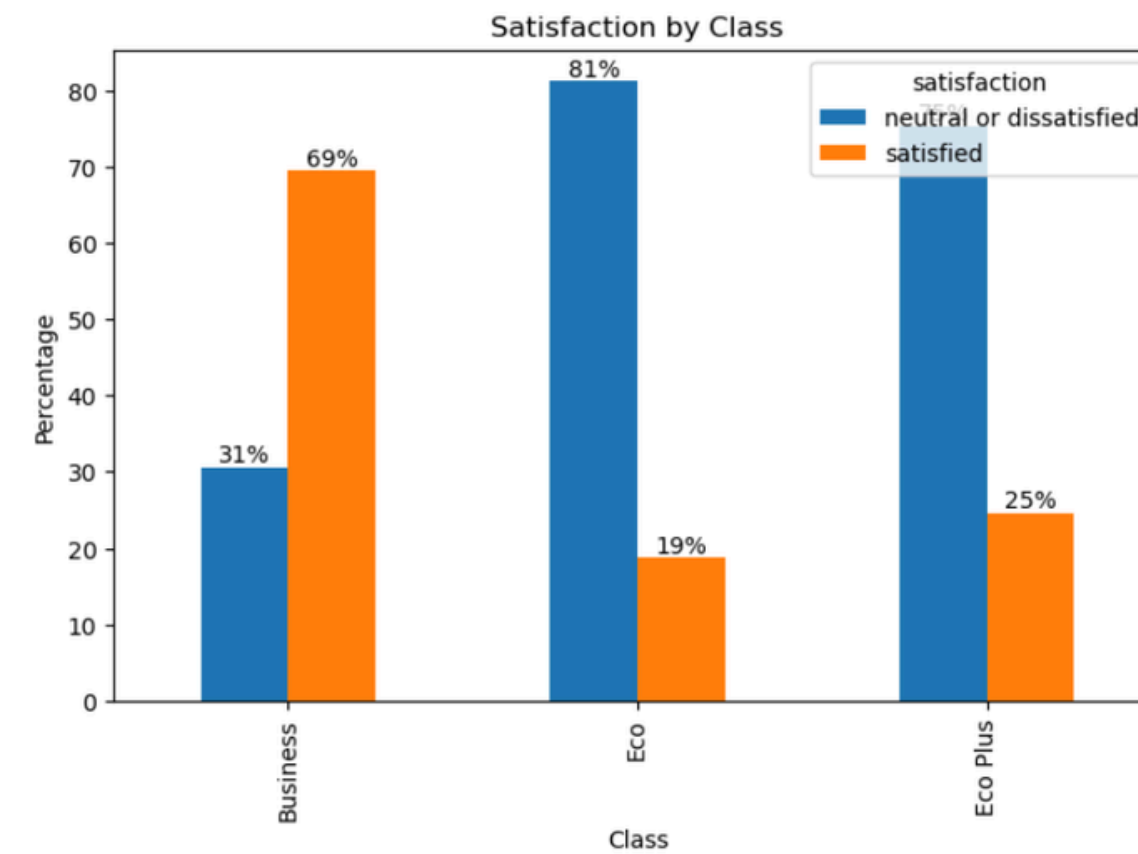
This distribution is important because travel class may be associated with passenger satisfaction and could therefore represent a relevant predictive feature.



Passenger Satisfaction by Class

Travel class shows a strong relationship with satisfaction.

Passengers traveling in Business Class are generally more satisfied than those traveling in Economy Class, highlighting the impact of service quality and travel conditions.

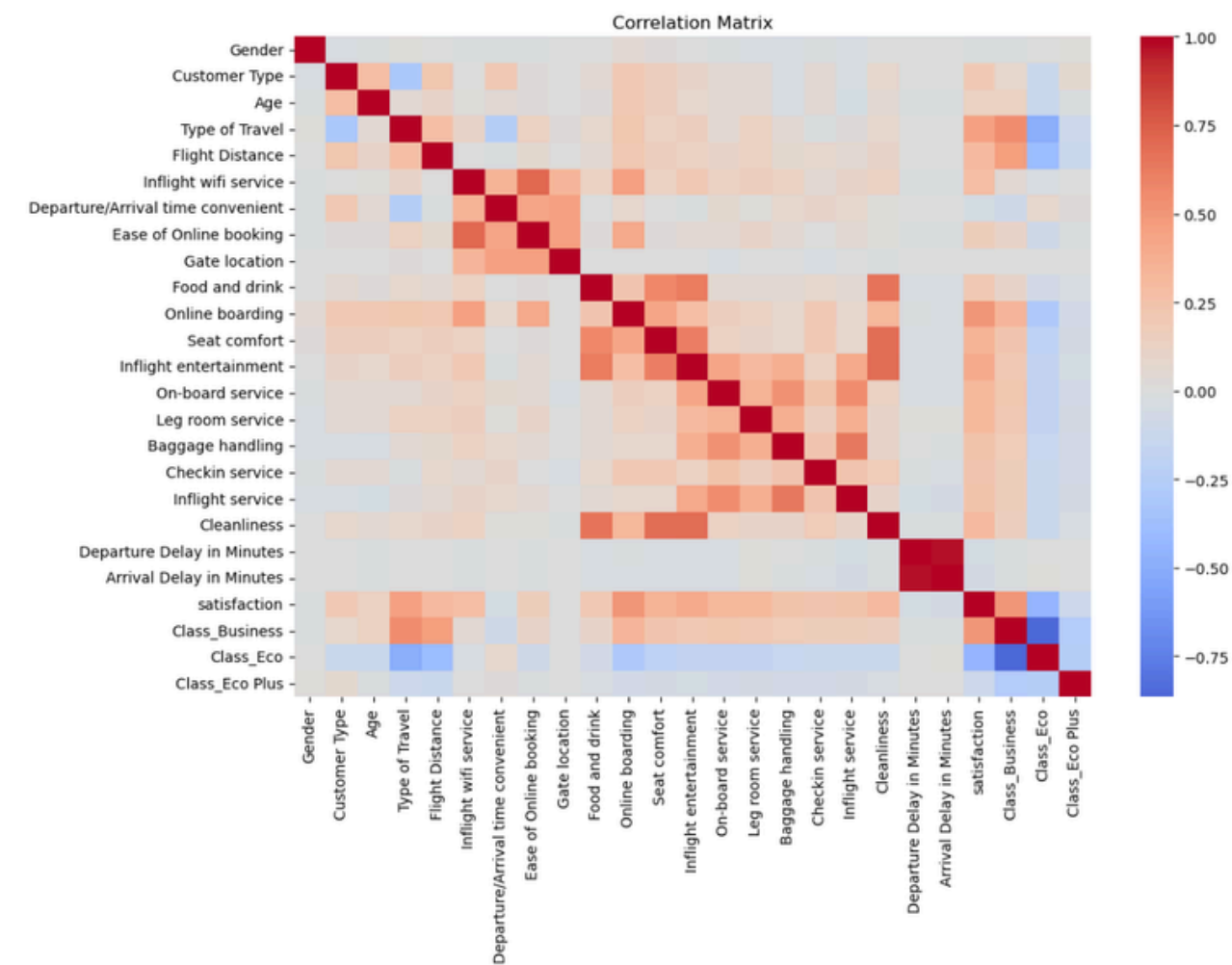


Correlation Analysis

The correlation analysis reveals several relationships between the numerical variables.

In particular, Departure Delay and Arrival Delay exhibit a strong positive correlation, which is expected given the nature of flight operations.

Although this introduces a degree of redundancy, both variables were retained as they represent different stages of the travel experience: the departure phase and the actual arrival outcome. In this context, the potential issue of multicollinearity is less critical for tree-based models, which are generally robust to correlated features.



Feature Selection

Methods

- Chi-Square Test
- Mutual Information

Main Findings

- Online Boarding
- Inflight WiFi Service
- Type of Travel
- Travel Class
- Seat Comfort

These features emerged as the most informative predictors of passenger satisfaction.

Observation

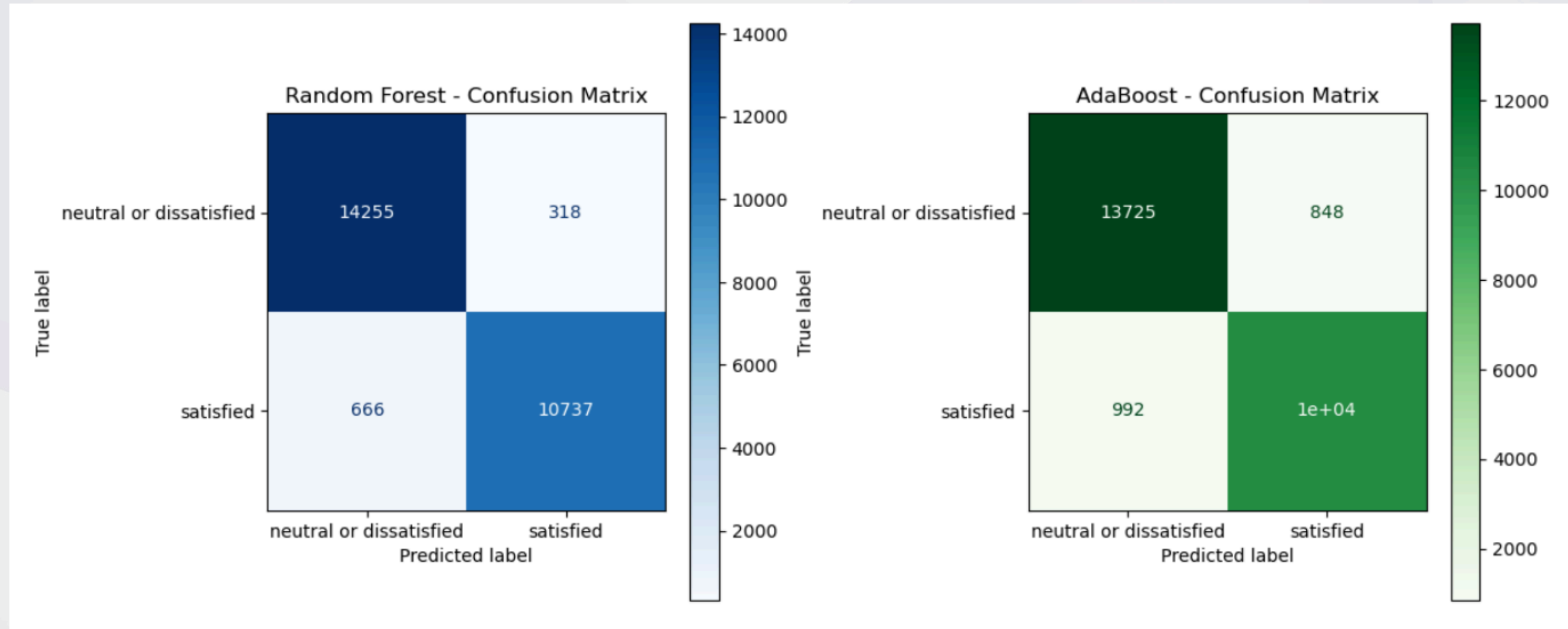
The selected features are consistent with the insights obtained during the EDA and later confirmed by the Random Forest feature importance analysis.

Model Comparison

Key Findings

- All machine learning models outperformed the DummyClassifier baseline.
- Models trained on the full feature set achieved better results than those trained on the reduced feature set.
- Feature selection slightly decreased performance, suggesting that some seemingly weak features still provide useful complementary information.
- Random Forest and AdaBoost outperformed Logistic Regression in cross-validation.
- Ensemble methods were better able to capture the complex and non-linear relationships underlying passenger satisfaction.

Model Evaluation Results



Key Findings

- Both models achieved strong classification performance across all evaluation metrics.
- Random Forest consistently outperformed AdaBoost in terms of accuracy, precision, recall, and F1-score.
- The Random Forest confusion matrix shows fewer overall misclassifications.
- Most errors are concentrated in borderline cases where passenger characteristics are less clearly associated with a satisfaction class.
- Overall, the results suggest that passenger satisfaction can be predicted effectively, although some overlap between classes remains.

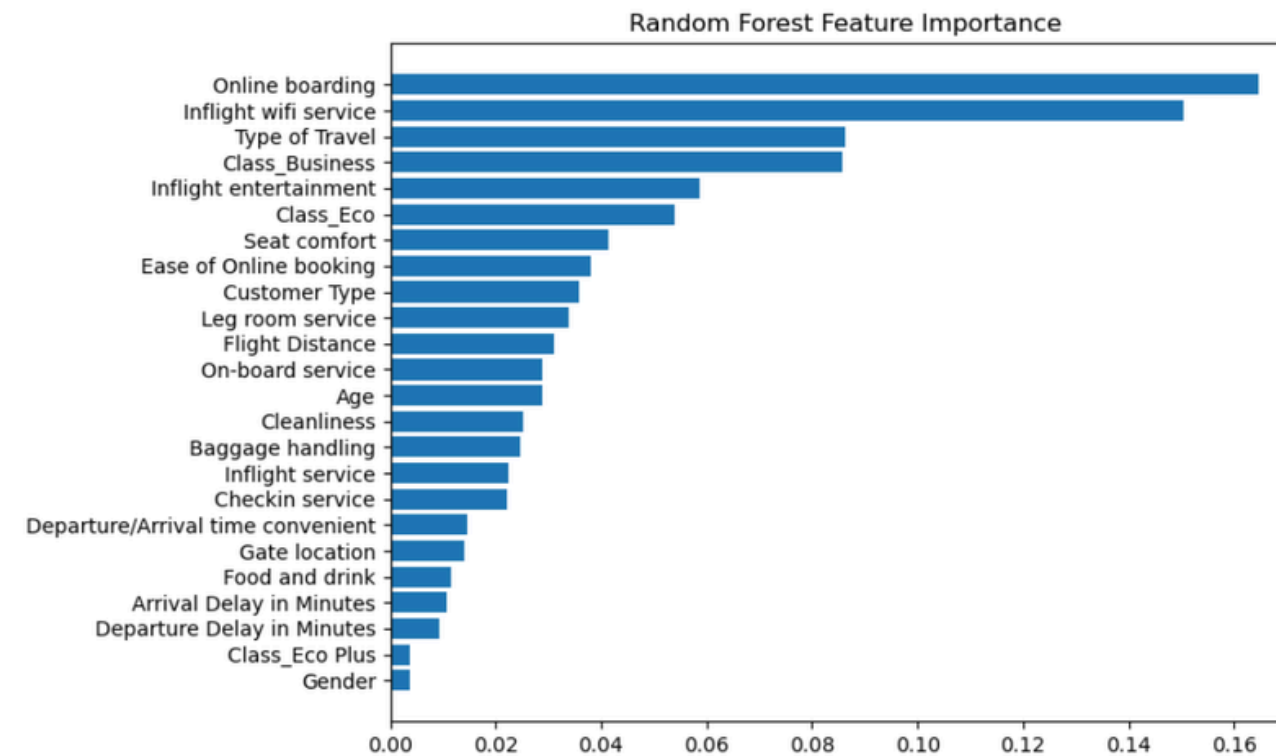
Feature Importance

Key Findings

- Customer experience features are the strongest predictors of satisfaction.
- Online Boarding and Inflight WiFi Service are the most influential variables.
- Type of Travel and Travel Class also play a major role in the prediction process.
- Comfort-related features (e.g., seat comfort, inflight entertainment, and leg room service) contribute substantially to model performance.
- Demographic variables and delay-related features have a relatively limited impact.

Business Insight

Passenger satisfaction is driven primarily by the quality of the travel experience rather than by demographic characteristics or flight delays.



Conclusions

Key Findings

- Service quality features emerged as the strongest predictors of passenger satisfaction.
- All machine learning models significantly outperformed the DummyClassifier baseline.
- Ensemble methods (Random Forest and AdaBoost) achieved better results than Logistic Regression, highlighting the importance of non-linear relationships and feature interactions.
- Random Forest delivered the best overall performance in terms of accuracy, robustness, and interpretability.
- Feature selection slightly reduced predictive performance, suggesting that seemingly weak features still provide useful complementary information.
- Both Random Forest and AdaBoost showed strong generalization ability and high stability across different random seeds.

Final Takeaway

A structured machine learning pipeline can effectively predict passenger satisfaction, with Random Forest emerging as the most suitable model for balancing predictive performance, robustness, and practical usability.

Project Impact

Business Insight:

Passenger satisfaction is primarily driven by service quality rather than demographic characteristics.

Practical Implication:

Investments in online boarding services, WiFi quality, seat comfort, and inflight entertainment are likely to produce the greatest improvements in customer satisfaction.

Best Model:

Random Forest